Dinesh Prabhakaran

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.35
Global Public School, Kota	Class XII (CBSE)	2011	73.40%
Faculty Higher Secondary School, Guwahati	Class X (CBSE)	2009	85.80%

SCHOLASTIC ACHIEVEMENTS

Co-authored a paper at ASME IMECE India 2026 on 'Liquid Ammonia Combustion Modelling in a CI Engine' in Sep 2026

Published a book chapter on 'Combustion of Ammonia Blends' with Springer Nature in Sept 2025

Presented a paper in 48th FMFP Conference, BITS Pilani on 'Combustion Mechanism for Lemon Peel Waste Oil' in Dec 2021

Presented a poster at National Symposium 'Shaping the Energy Future: Challenges and Opportunities (SEFCO)' in Aug 2021

Received silver medal during M. Tech for securing 1st rank in the Department of Climate Change in Jun 2021

Presented a poster at 'National Conference on Materials for Energy Conversion and Storage' in VIT Chennai on 20th Mar 15

POSITIONS OF RESPONSIBILITY

Teaching Assistant for NPTEL Course on Sustainable Energy Technology from 2023 to 2025

Social media coordinator for the Department of Climate Change, IITH from 2021 to 2023

PhD Mentor at Sunshine, the counselling cell of IITH in academic year 2022-23

SKILLS

Application Software : CHEMKIN, OpenSMOKE, AutoCAD, MATLAB, Solid Works, ERP, Microsoft Office Suite

Programming Languages : Python, C, FORTRAN

SELECT COURSES

Combustion Technology, Computational Fluid Dynamics, Introduction to AI & ML, Climate Change Sciences & Mitigation

PROJECTS

PhD Thesis - Advisers: Dr. Sayak Banerjee and Prof. Raja Banerjee

Chemical Kinetic Study of Ammonia-Diesel and Ammonia-Bio-diesel Blends for CI Engine Operation

- Constructed and validated a semi - detailed ammonia - diesel combustion chemical mechanism

- Investigated the combustion and emission characteristics of ammonia - diesel blends in CI engine systems

- Assemble and validate a semi - detailed ammonia - bio-diesel combustion chemical mechanism

- Reduce the two semi-detailed mechanisms for complex turbulent combustion modelling

- Compare the combustion and emission characteristics of diesel, ammonia-diesel and ammonia-biodiesel blends

- Suggest optimum blend ratios of alternate fuels in CI engines to maximise performance and minimise emissions

Master's Thesis - Adviser: Dr. Sayak Banerjee

Development of a Reduced Combustion Kinetic Mechanism for Lemon Peel Oil (LPO) as a Jet-fuel

- Researched the potentiality of replacing fossil fuel based jet fuel by a bio-fuel derived from waste lemon peel

- Studied preliminary combustion characteristics of LPO using chemical kinetic modelling software

- Reduced a chemical kinetic combustion mechanism for lemon peel oil surrogates

- Compared the combustion properties using the reduced mechanism with those using the detailed mechanism

- Performed preliminary flat flame experiments geared towards future experimental validation

Bachelor's thesis - Adviser: Dr. R. Pratibha Nalini

Nano Structured Solar Cell

- Fabricated an innovative solar cell model to improve its efficiency over that of already existent models

- Created a nano-scale thin film of silicon carbide on silicon substrate by co-sputtering using silicon and carbon targets

- Studied the properties of the new model by various characterization tests like X-ray diffraction and atomic force microscopy

- Found that the model showed improved behavior in terms of band gap and refractive index

EXTRACURRICULAR

Participated in 'SIC Learn in Nippon' - a two-week educational and cultural exchange program with Suzuki Motor Corporation, Japan sponsored by Suzuki Innovation Center, IITH in Sept 2023

WORK EXPERIENCE

Hyundai Mobis India Ltd., Inventory Management & Internal Audit (Chennai)

- Managed inventory for after-sales parts and accessories of Hyundai cars for East India region

- Planned stock requirement based on demand analysis & forecasts by understanding market trend

- Arranged emergency/critical order parts for dealers all over India on priority and prepared performance reports

- Identified internal control weaknesses and delivered value-added recommendations to management

- Suggested improvements - like better scrap management & efficient procurement of packing materials

OTHER DETAILS

- **Hobbies** - Art and Craft, Reading, Travelling and Cooking; - **Languages known** - English, Tamil, Hindi, Assamese